High-Recall Deep Learning: A Gated Recurrent Unit Approach to Bank Account Fraud Detection on Imbalanced Data

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Abstract: Fraud detection in financial services is challenged by severe class imbalance, where fraudulent events are rare. This study provides a rigorous comparison of classical machine learning models—Logistic Regression, SVM, Random Forest, and LightGBM—and a Gated Recurrent Unit (GRU) deep learning model on a large-scale, imbalanced bank account fraud dataset. Our findings reveal a dramatic performance divide. All classical models, including the widely-used tree-based ensembles, proved ineffective at the primary detection task, with fraud recall scores below 7%. In stark contrast, the GRU network successfully identified the vast majority of fraudulent cases, achieving over 78% recall on the minority fraud class, albeit at the cost of very low precision. The results demonstrate that the GRU offers a viable, high-recall solution where classical models fail. This highlights a critical strategic choice for financial institutions: adopt a high-recall/high-alert detection framework or use conventional models that allow most fraud to go undetected. The approach is broadly applicable across financial domains—including banking and insurance—where rare-event detection is critical.

Keywords: *Supervised Learning, Fraud Detection, Class Imbalance, Deep Learning, Gated Recurrent Unit (GRU), Precision-Recall Trade-off, Tabular Data.*

I. Introduction

Fraudulent activity in financial services poses a persistent and evolving threat, but it is not limited to transaction-level scams like credit card fraud. A particularly costly and growing danger is bank account opening fraud, where individuals use stolen or synthetic identities to create new accounts. Once established, these accounts serve as conduits for money laundering, unauthorized credit access, and other forms of financial crime. The scale of this problem is substantial, with industry reports estimating billions in annual losses [1-2], in addition to the significant operational and reputational risks that financial institutions incur when fraudulent accounts slip through their onboarding systems.

Beyond banking, these detection challenges have direct implications for the insurance industry. Insurers are increasingly exposed to cyber fraud, claims fraud, and identity-related schemes that present the same imbalanced data challenges found in financial services. By adapting the type of fraud detection frameworks explored in this study, insurance carriers could strengthen underwriting, improve claims validation, and better protect themselves and their policyholders against rapidly evolving threats [3,4].

Historically, the first line of defense against new account fraud has been traditional detection frameworks. These systems rely heavily on static, hard-coded rules, such as flagging mismatched addresses or unusual identity combinations. While these frameworks are easy to interpret, their core challenge lies in their rigidity. Fraud tactics evolve quickly, and these rule sets often fail to adapt, leading to two critical failures: they struggle to catch novel fraudulent patterns while simultaneously overwhelming investigators with a high volume of false positives. The sheer volume and velocity of transactions in the modern digital economy render these manual or static systems computationally infeasible, creating an urgent need for automated, scalable solutions. This lack of adaptability necessitates a more dynamic approach.

To overcome these limitations, machine learning (ML) has been increasingly adopted for its ability to learn subtle, complex patterns from heterogeneous data [2,5]. These models are critical components of larger financial technology (FinTech) platforms, often deployed on cloud infrastructure where high throughput and low-latency predictions are essential for real-time decision-making. Unlike static rules, ML models can analyze demographic details, digital footprints, and behavioral indicators to offer more flexible and adaptive fraud detection. However, applying ML in this domain introduces its own formidable challenge: severe class imbalance. With fraudulent applications constituting a very small fraction of all cases, models trained on such skewed distributions can achieve high overall accuracy while failing at their primary task of identifying the rare fraudulent accounts. This can create a dangerous illusion of performance, making class imbalance the central obstacle to effective ML-based fraud detection [7].

To systematically investigate how different modeling approaches perform under the strain of severe class imbalance, this study provides a rigorous comparative analysis. Our research plan is centered on the publicly available Bank Account Fraud (BAF) dataset, a large-scale benchmark from the NeurIPS

2022 track that mimics the real-world account-opening process. Using this dataset, our study will:

- Evaluate a spectrum of supervised algorithms, ranging from interpretable linear classifiers and tree-based ensembles to a recurrent neural network.
- Analyze performance by focusing on minority-class metrics, moving beyond simple accuracy to assess how well each model identifies the rare but critical fraud cases.
- Investigate the trade-offs between precision and recall, which correspond directly to the operational risks of managing false positives versus the financial risk of missing fraud.

Through this structured evaluation, this paper aims to provide clear insights into which modeling strategies are best suited for the nuanced and high-stakes task of bank account fraud detection.

II. Methods

A. Dataset and Preprocessing

This study utilizes the Bank Account Fraud (BAF) dataset, a large-scale, privacy-preserving benchmark for tabular fraud detection released at NeurIPS 2022. We use the "Base" variant, which contains approximately one million synthetic bank account applications generated from a real-world, anonymized source. Each record is described by a range of features and includes a binary label to indicate if the application was fraudulent. A defining characteristic of this dataset is its severe class imbalance, with fraudulent applications constituting only 1.10% of the total records.

The features provide a comprehensive snapshot of each application, containing signals related to the applicant's profile and behavior. The numeric variables capture demographic and financial indicators such as the applicant's age and income, alongside an internal credit risk assessment. We also included behavioral velocity metrics, which track the rate of application activity over various time windows, from hours to weeks. The categorical attributes describe the applicant's professional and housing status, the type of device used for the application, and whether the request originated from a foreign country.

From the available data, two features were excluded from all analyses. The first, a variable indicating the duration at a previous address, was removed due to a high volume of missing values. The second, a temporal indicator for the application month, was dropped to prevent potential data leakage from distributional shifts over time.

To prepare the data for modeling while ensuring a fair evaluation, we first split the full dataset into training (80%) and testing (20%) partitions. This was performed using stratified sampling to preserve the original distribution of the minority fraud class in both sets. All subsequent preprocessing steps were fitted exclusively on the training data to avoid contaminating the test set. For numeric features, missing values were imputed using the median, and the features were then normalized via z-score standardization. For categorical features, missing values were replaced with a placeholder string, and the features were then transformed using one-hot encoding. Finally, the feature names were extracted post-transformation to ensure that any subsequent analyses, such as variable importance plots, would have interpretable labels.

B. Machine Learning Models

To evaluate the effectiveness of classical supervised methods in detecting fraudulent activity, we implemented four representative algorithms spanning linear, kernel-based, and tree-based models. All models were trained on the same processed dataset, and hyperparameters were tuned using stratified cross-validation with the weighted F1 score as the primary criterion.

Logistic Regression

We first established a benchmark using Logistic Regression [7], a straightforward linear classifier. The model estimates the probability of fraud as a logistic transformation of a weighted combination of features. Despite its simplicity and inability to capture nonlinear relationships, Logistic Regression remains valuable due to its interpretability and computational efficiency, especially in domains where transparency is critical. During tuning, we varied the regularization strength (C), the optimization solver (liblinear and lbfgs), and the treatment of class imbalance via balanced class weights. This configuration served as our baseline against which more sophisticated models were compared.

Support Vector Machine (SVM)

Support Vector Machines were included to extend the analysis beyond linear separation [8,9]. Through constructing hyperplanes that maximize the margin between classes, SVMs can model both linear and nonlinear boundaries when paired with kernels. In our experiments, we tested linear and radial basis function (RBF) kernels, adjusting parameters such as the penalty term (C), kernel scaling (gamma), and class weighting strategies. Although computationally heavier than Logistic Regression, SVMs offer the ability to discover more complex decision surfaces, which can be valuable in fraud detection where patterns are subtle and nonlinear. This higher computational cost is due to the algorithm's training complexity, which can be between $O(n^2)$ and $O(n^3)$ in the worst case, making training on datasets of this scale particularly demanding.

Tree-based Models

In addition to linear and kernel methods, we examined tree-based ensemble algorithms. Decision-tree ensembles are particularly appealing for fraud detection because they naturally capture nonlinear feature interactions and handle mixed data types. Within this family, two complementary strategies dominate: bagging, which reduces variance by training multiple trees on bootstrapped data and averaging their predictions, and boosting, which builds trees sequentially, each correcting the errors of the previous stage. Random Forest exemplifies bagging, while Light Gradient Boosting Machine (LightGBM) represents boosting.

Random Forest: Bagging Techniques

Random Forest constructs an ensemble of decision trees [10], each trained on a randomly resampled subset of the training data and a random subset of features. The model aggregates predictions across trees through majority voting, which reduces overfitting and enhances robustness compared to a single tree. Key hyperparameters tuned included the number of estimators, tree depth, and minimum sample requirements for splits and leaves. To address class imbalance, we also explored balanced class weighting during training.

Light Gradient Boosting Machine: Boosting Techniques

LightGBM serves as the boosting counterpart [11,12]. Unlike Random Forest's parallel bagging strategy, LightGBM builds trees sequentially, with each new tree focused on reducing the residual errors of the prior ensemble. Its efficiency arises from histogram-based feature binning and leaf-wise growth, enabling faster training and scalability to large datasets. We optimized parameters such as the number of boosting rounds, learning rate, maximum depth, number of leaves, and minimum child samples. LightGBM combines predictive strength with computational speed, making it a strong contender in fraud detection tasks.

C. Deep Learning Models

Although linear methods, kernel-based algorithms, and tree ensembles remain the core toolkit for fraud detection, deep learning has the capacity to capture richer dependencies across heterogeneous features. Neural networks, particularly architectures designed for sequences, can be adapted to tabular fraud detection by treating categorical inputs as tokenized embeddings and modeling interactions through hidden states.

In this study, we deliberately limited our scope to a single deep learning model: the Gated Recurrent Unit (GRU) [13]. Unlike previous work that evaluates a suite of neural architectures (e.g. transformers, LSTMs, and GRUs), our objective was not to exhaustively benchmark deep learning families but rather to assess whether a compact, sequence-oriented model could add predictive value when compared against classical baselines. GRUs were chosen because they provide a balance between expressive power and computational efficiency, offering gated mechanisms to handle long-term dependencies without the overhead of more complex models such as LSTMs or BERT-style transformers.

Gated Recurrent Unit (GRU)

The GRU model was implemented in PyTorch. The network architecture was constructed using PyTorch's nn.Module. Categorical features were first converted to dense vectors using an nn.Embedding layer, while numeric features were processed as a single tensor. These two sets of features were then concatenated into a single input vector for each data sample, which was treated as a sequence of feature groups. To provide a structured input for the recurrent architecture, the tabular features were ordered thematically rather than randomly. The sequence was arranged to flow from stable, long-term attributes (e.g., demographic and financial data) to dynamic, short-term indicators (e.g., behavioral velocity and session data). This approach aimed to mimic a logical flow of information analysis. This combined tensor was fed into a multi-layer nn.GRU module, whose final hidden state was passed to an nn.Linear layer for classification. Continuous variables were first normalized and mapped into dense representations, while categorical variables were embedded into trainable low-dimensional vectors [13]. These representations were concatenated and processed sequentially by the GRU layer, whose update and reset gates modulated information flow and memory retention. The final hidden state was passed through dropout regularization and a fully connected layer to produce fraud predictions.

To handle the pronounced class imbalance, we applied a weighted cross-entropy loss with class weights inversely proportional to class frequencies. The model was optimized using Adam, and hyperparameters were tuned via random search across 10 candidate settings. Search spaces included embedding size (150–250), hidden dimension (256–768), learning rate (1-4–1-3), and number of epochs (5–10). Validation was stratified to preserve class distribution, and the weighted F1 score was used as the primary selection metric.

By focusing on GRUs alone, we created a controlled evaluation setting that highlights whether sequence-inspired neural networks can complement or surpass the performance of more traditional machine learning techniques in the context of fraud detection.

D. Evaluation Metrics

In fraud detection, the overwhelming skew toward legitimate cases means that accuracy alone can be misleading. A model that predicts "legitimate" for nearly every instance could achieve high accuracy, yet completely fail at identifying fraudulent behavior. To avoid this pitfall, we adopted a set of complementary evaluation measures that emphasize minority-class performance [14,15].

Our analysis centered on four widely used metrics: precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (AUC-ROC). Precision captures how often the cases flagged as fraud are truly fraudulent, while recall measures how many of the actual fraud cases the model successfully detects. In this application, recall carries particular weight: missing a fraudulent account (false negative) poses significant financial and security risks. Precision, however, is equally important in operational contexts, since excessive false positives can overwhelm analysts and erode trust in the detection system.

The F1-score was used as our main selection criterion because it integrates both precision and recall into a single statistic, providing a balanced measure of performance under class imbalance. AUC-ROC served as a complementary benchmark, evaluating the model's ability to discriminate between classes across different thresholds.

This framework was applied consistently across all algorithms—both classical machine learning models and the GRU neural network—ensuring that results were directly comparable and fairly assessed under the conditions of highly imbalanced classification.

III. RESULTS

A. Hyperparameter Selection

For each model, we conducted systematic hyperparameter searches. As systematic reviews of automated machine learning suggest, hyperparameter optimization is a critical step for minimizing model bias and variance, ensuring that a model's complexity is well-suited to the data [16,17]. Therefore, following common practice for computationally intensive tasks [18-20], particularly with large datasets, candidate settings were evaluated on a stratified subset of 10,000 training samples rather than the full dataset to ensure tractability. The primary metric for this evaluation was the weighted F1 score, which balances the trade-off between precision and recall.

For the classical machine learning models, hyperparameters were tuned via grid search. In Logistic Regression, we varied the regularization strength (C), tested multiple solvers (liblinear and lbfgs), and compared models with and without class weighting. For Random Forest, the grid included the number of trees, maximum depth, minimum node size, and class weighting. LightGBM was tuned over the number of boosting rounds, learning rate, maximum depth, leaf count, minimum child samples, and class weighting.

Support Vector Machines are particularly expensive to train at scale, which further justified the subset-based approach. The SVM grid covered kernel type (linear vs. RBF), regularization parameter (C), kernel coefficient (γ), and class weighting. The best-performing configuration was an RBF kernel with C=10, γ=scale, and no class weighting.

The Gated Recurrent Unit (GRU) network, due to the higher cost of deep learning optimization, was tuned using random search over embedding dimensions, hidden layer sizes, learning rates, and training epochs. Due to the significant computational expense of training the GRU on a large-scale dataset, a random search limited to 10 trials was conducted to find a strong baseline configuration. While a more exhaustive search might yield further incremental improvements, this tuning was sufficient to establish whether the GRU architecture could fundamentally overcome the recall limitations of classical models. The best configuration consisted of an embedding dimension of 186, hidden size of 733, learning rate of 4.48×10^{-4}, and 8 epochs.

The final parameter choices for each algorithm are summarized in Table 1, ensuring fair comparisons under imbalanced classification conditions.

TABLE1: SUMMARY OF OPTIMIZED HYPERPARAMETER CONFIGURATIONS

Model	**Key Hyperparameters**
Logistic Regression	C=100, solver=lbfgs, penalty=l2, class_weight=None
SVM	kernel = rbf; C = 10; gamma = scale; class_weight = None
Random Forest	n_estimators=200, max_depth=10, min_samples_leaf=4, min_samples_split=10, class_weight=Balanced
LightGBM	n_estimators=100, lr=0.1, max_depth=-1, num_leaves=31, min_child_split=10, class_weight=None
GRU	Embedding dimension = 186; Hidden units = 733; Learning rate = 4.48×10^{-4}; Epochs = 8

B. *Model Performance*

At first glance, the weighted average metrics in Table 2 suggest that the four classical models are nearly perfect and perform almost identically, each achieving a weighted F1-score of approximately 0.98. This extremely high score, however, is an "illusion of accuracy" created by the severe class imbalance in the dataset. These metrics are dominated by the models' excellent performance on the non-fraud class, which constitutes over 98% of the data, and therefore do not reflect true fraud detection capability. In contrast, the Gated Recurrent Unit (GRU) model appears to be a weaker performer with a weighted F1-score of 0.897. This lower score indicates that the GRU makes more errors on the majority class, hinting at a fundamentally different classification strategy compared to the classical models.

TABLE 2: AGGREGATE TEST-SET PERFORMANCE METRICS

Model	**Weighted Average**			**AUC**
	Precision	Recall	F1-Score	
Logistic Regression	0.98	0.99	0.9836	0.8487 (0.8401–0.8566)
SVM	0.98	0.99	0.9835	0.7465 (0.7361–0.7566)
Random Forest	0.98	0.99	0.9838	0.8551 (0.8476–0.8630)
LightGBM	0.98	0.99	0.9834	0.8144 (0.8052–0.8234)
GRU	0.99	0.83	0.8971	0.8800 (0.8300–0.8970)

A more meaningful story is told by the Area Under the ROC Curve (AUC), which measures a model's ability to distinguish between classes across all decision thresholds and is less sensitive to class imbalance. According to this metric, the GRU (0.880) and Random Forest (0.855) are the clear top performers, demonstrating the strongest underlying ability to learn the patterns that separate fraudulent applications from legitimate ones. The Support Vector Machine (SVM), despite its high weighted F1-score, reveals a critical weakness with the lowest AUC by a significant margin (0.747), indicating it is the poorest model at separating the two classes.

Synthesizing these two perspectives reveals the classic trade-off in fraud detection. The high weighted F1-scores of the classical models reflect a conservative strategy that excels at identifying non-fraud cases, while the GRU's lower weighted F1-score is a consequence of a more aggressive strategy that flags more potential fraud at the cost of more false positives. Ultimately, the AUC is the most important metric for evaluating performance on this task, as it measures true discriminative power. Based on this, the GRU and Random Forest are the most effective and promising models developed in this study.

Table 3 shifts the focus from aggregate performance to the single most important aspect of the study: how well each model detects the rare but critical fraudulent cases. By isolating the precision, recall, and F1-score for the minority (fraud) class, the table reveals the dramatic and practical differences in each model's strategy, a nuance that was hidden in the weighted averages of Table 2.

TABLE 3: MINORITY CLASS (FRAUD) DETECTION METRICS

Model	**Precision (Fraud)**	**Recall (Fraud)**	**F1-Score (Fraud)**
Logistic Regression	0.46	0.01	0.01
SVM	0.16	0.01	0.02
Random Forest	0.24	0.06	0.10
LightGBM	0.19	0.06	0.09
GRU	0.049	0.781	0.092

The results show a clear failure of the linear models in this context. Both Logistic Regression and the SVM exhibit a recall of just 0.01, meaning they successfully identify only 1% of all fraudulent applications. While their precision is moderate to low, their inability to catch fraudsters makes them operationally ineffective for this task. Their strategy is so conservative and focused on avoiding mistakes on the non-fraud class that they effectively ignore the primary threat. The tree-based ensembles, Random Forest and LightGBM, offer a modest improvement, increasing fraud recall to 0.06. However, they still fail to detect the vast majority (94%) of fraudulent activity, resulting in low F1-scores of around 0.10 and proving them to be insufficient solutions.

The GRU model demonstrates a completely different and far more compelling strategy. It achieves an outstanding fraud recall of 0.781, successfully identifying over 78% of all fraudulent cases—a massive leap in performance that no other model approaches. This high recall, however, comes at the cost of a very low precision of 0.049. In practical terms, this means that while the GRU is highly effective at catching fraudsters, it also generates a large volume of false alarms; for every 100 cases it flags as fraudulent, only about 5 are truly fraudulent. This trade-off is fundamental in fraud detection: the GRU is the only model that solves the primary problem of catching fraud, but it does so by accepting a high operational cost of investigating false positives.

IV. Discussion

A. *Interpretation of Findings*

This study benchmarked five supervised approaches on a large, imbalanced bank account fraud dataset, revealing a stark divide in their effectiveness. The primary finding is the dramatic difference between the GRU network's high-recall strategy and the conservative, low-recall strategies of all four classical machine learning models. While the tree-based ensembles are often favored for such tasks, our results show they were only marginally better than linear models and failed to adequately address the core challenge.

The Logistic Regression and SVM models exemplified the classic weakness of linear classifiers on this problem. Despite high weighted F1-scores, their recall on the fraud class was negligible (0.01), meaning they missed 99% of true fraud cases. A McNemar's test confirmed their strategic similarity, showing no statistically significant difference in their predictions ($p > 0.05$). This proves they are unsuitable for this task, as they prioritize avoiding false positives to the point of ignoring the actual threat.

Theoretically, the ensemble trees (LightGBM and Random Forest) should capture complex, non-linear patterns. In practice, however, they offered only a minor improvement, increasing fraud recall to just 0.06. While statistically different from the linear models, they still allowed 94% of fraud to go undetected, demonstrating that a more powerful approach is necessary. They do not, as is often assumed, represent a balanced or effective solution in this high-imbalance scenario.

In stark contrast, the GRU presented a breakthrough. It was the only model to solve the recall problem, successfully identifying over 78% of fraudulent applications. This high recall came at the cost of very low precision, resulting in a high rate of false positives. This unique performance profile was confirmed to be statistically significantly different from all classical models ($p < 0.001$). The GRU's ability to learn from the feature sequences without extensive engineering proved to be the key to overcoming the central challenge of the dataset.

B. *Practical Implications*

From an applied perspective, these findings suggest a clear hierarchy of models based on the business priority of catching fraud. Logistic Regression remains useful only when interpretability is the sole concern and fraud detection is a secondary goal. The tree-based ensembles, particularly Random Forest, represent an incremental improvement but do not meet the recall demands of a high-stakes fraud system.

The GRU, despite its high false positive rate, is the strongest candidate for production systems where the primary objective is to minimize the financial and security losses from missed fraud. Its deployment would necessitate an operational workflow designed to manage and triage the resulting alerts, for instance, by using a secondary model or a tiered review system. Furthermore, the choice of model carries significant implications for computational resources and system design. The classical models vary in efficiency: Logistic Regression offers maximum speed and low overhead, making it an efficient baseline, while the SVM with an RBF kernel proved computationally expensive to train at scale, a common challenge for kernel-based methods on large datasets. Of the classical approaches, LightGBM stands out for its computational performance, leveraging histogram-based techniques to enable fast training and scalability. In contrast, the GRU network, as is typical for deep learning models, presented the highest computational demand, requiring longer training cycles and specialized hardware to capture the complex patterns necessary for its high recall performance. This establishes a critical engineering trade-off: the superior detection capability of the GRU comes at the cost of greater computational investment, a factor that must be weighed against the business cost of missed fraud when architecting a production-level detection system. The crucial takeaway is that the cost of managing the GRU's false positives is a more manageable business problem than the cost of missing 94-99% of fraud, as the other models did.

Fraud detection remains a critical challenge across financial services, encompassing both banking and insurance. In banking, fraudulent accounts and suspicious transactions expose institutions to direct financial losses and systemic risks, while also undermining customer trust and the integrity of the broader financial system. In insurance, fraud takes the form of misrepresented risks, inflated or false claims, and emerging exposures such as cyber incidents—all of which directly impact actuarial assessments of loss ratios, pricing adequacy, and portfolio stability. Despite these domain-specific differences, both industries face the same underlying difficulty: rare but costly fraud events hidden within massive volumes of legitimate data. Advanced detection frameworks, such as the GRU approach explored in this study, provide a pathway to address these challenges by capturing minority-class patterns that traditional models miss.

C. *Limitations and Future Works*

This analysis is subject to several constraints. First, the synthetic dataset may not fully capture the evolving strategies of real-world fraudsters. Second, our deep learning exploration was limited to a GRU; extending the comparison to transformer-based architectures could reveal further benefits. Third, we acknowledge that our hyperparameter search for the GRU model was constrained by computational resources. Future work could employ more advanced optimization techniques, such as Bayesian optimization over a larger search space, to potentially further enhance model performance. Fourth, we did not directly model the asymmetric business costs of false negatives versus false positives. Finally, our evaluation was static, whereas a real-world deployment would require adaptive systems capable of handling concept drift over time. Future research should therefore focus on developing cost-sensitive, adaptive models tested in production environments, with particular attention to their application across financial services such as banking and insurance. Both domains share common challenges of rarity, high costs of missed fraud, and rapidly evolving attack strategies, making them ideal testbeds for advanced approaches. By tailoring such models to the operational realities of these industries, researchers can deliver not only technical improvements but also meaningful business impact in reducing fraud-related losses and strengthening financial resilience.

D. Summary

This study provides a rigorous comparative evaluation of classical machine learning and deep learning models for the task of bank account opening fraud detection on a large, highly imbalanced benchmark dataset. Our findings demonstrate that traditional approaches, including Logistic Regression, SVM, and even tree-based ensembles like Random Forest and LightGBM, proved largely ineffective at the primary task, failing to identify the vast majority of fraudulent cases with fraud-class recall scores near or below 6%. In stark contrast, a Gated Recurrent Unit (GRU) network successfully addressed this core challenge by identifying over 78% of fraudulent applications, a breakthrough in performance that came with the significant trade-off of very low precision and a high volume of false alarms. The study concludes that the GRU is the only tested model viable for minimizing fraud losses, highlighting a critical strategic choice between a high-recall/high-alert approach and other methods that inadequately address the problem, thereby emphasizing the need to align model selection with specific business risk tolerance and operational priorities.

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